## Shu-Hsien Cho


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**Research Interests** Statistical Genetics and Genomics, Rare Variant Association Test and Variant Prioritization in Biobank-scale Data, Causal Mediation Analysis, Survival Analysis.

**Education**

**University of Texas MD Anderson Cancer Center UTHealth Graduate School of Biomedical Sciences**, Houston, TX

Ph.D., Quantitative Sciences (Biostatistics Track) , *Expected:* May 2027

- Dissertation: *Deleterious Rare Variant Association Test and Variant Prioritization in Biobank-scale Data*
- Advisor: Chad D. Huff, Ph.D.

**National Chengchi University**, Taipei, Taiwan

B.S., Statistics , B.A., Economics (Double Major), *Certificate in Mathematical Finance*, *Certificate in Actuarial Statistics*, Jan 2017

**Research Experience**

**Graduate Research Assistant**
University of Texas MD Anderson Cancer Center UTHealth Graduate School of Biomedical Sciences
Supervisor: Chad D. Huff, Ph.D. Jul 2023 to now

- Developed statistical genetics methods for rare-variant association testing (VACANT, VACANT-M) and a variant-annotation scoring framework (SHARC) (see Working Papers).
- Contributed a secondary heterogeneity analysis to a large-scale genome-wide association study of stuttering (*Nature Genetics*, 2025; see Publications).

Supervisor: Jing Ning, Ph.D. and Yu Shen, Ph.D. May 2021 to Sep 2022

- Developed a novel adaptive information-borrowing framework for the Cox survival model to improve prognostic prediction for Triple-Negative Breast Cancer (TNBC).
- Quantified the heterogeneity between the target TNBC clinical dataset and a larger population cohort using both iterative maximum likelihood estimation (MLE) and estimating equations to calibrate the degree of information sharing.

**Research Assistant** Mar 2017 to Jul 2020
Institute of Statistical Science, Academia Sinica
Supervisor: Yen-Tsung Huang, MD, ScD

- Extended the causal mediation framework for Cox models (see Publications) to a semiparametric approach for semi-competing risks, enabling inference on pathway-specific cumulative hazards.
- Contributed statistical analysis to a randomized controlled trial on statins for post-cauterization liver cancer recurrence (Investigator: Dr. Yao-Chun Hsu).
- Organized a machine-learning study group and training program for the lab and a collaborating lab at the Institute of Molecular Biology.

**Journal Publications**

1. Polikowsky, H. G., Scartozzi, A. C., Shaw, D. M., Pruett, D. G., Chen, H.-H., Petty, L. E., Petty, A. S., Lowther, E. J., **Cho, S-H.**, Yu, Y., Mozaffari, S., Avery, C. L., Harris, K. M., Gordon, R. L., Beilby, J. M., Viljoen, K. Z., Jones, R. M., Huff, C. D., Highland, H. M., Kraft, S. J., and Below, J. E. "Large-scale genome-wide analyses of stuttering" *Nature Genetics* (2025) 57(8):1835-1847.

2. **Cho S-H.**, and Huang Y-T., "Mediation Analysis with Causally Ordered Mediators Using Cox Proportional Hazards Model" *Statistics in Medicine.* (2019) 38(9):1566-1581.

3. Hsu Y-C., Jun T., Huang Y-T., Yeh M-L., Lee C-L., Ogawa S., **Cho S-H.**, Lin J-T., Yu M-L., Nguyen M. H., and Tanaka Y., "Serum M2BPGi level and risk of hepatocellular carcinoma after oral anti-viral therapy in patients with chronic hepatitis B" *Alimentary Pharmacology & Therapeutics.* (2018) 48(10):1128-1137.

## Working Papers

1. **Cho, S-H.**, Bohlender, R. J., Yu, Y., and Huff, C. D. "VACANT: Variant Annotation Clustering AssociatioN Test." Manuscript in preparation.

2. **Cho, S-H.**, Bohlender, R. J., Yu, Y., and Huff, C. D. "VACANT-M: A Mixed-Model Extension of VACANT Accounting for Genetic Relatedness and Population Structure." Manuscript in preparation.

3. **Cho, S-H.**, Bohlender, R. J., Yu, Y., and Huff, C. D. "SHARC: Score Heterogeneity and Allele-type Recalibration for Coding-variants." Manuscript in preparation.

## Ongoing Projects

1. **Cho, S-H.** "Mahalanobis distance as a unifying power criterion for rare-variant association tests." In progress.

2. **Cho, S-H.** "A companion analysis extending the unifying power framework to exact tail-probability behavior for rare-variant association tests." In progress.

3. **Cho, S-H.** "Causal decomposition of MLB's Automated Ball-Strike Challenge System: umpire behavioral change versus algorithmic correction." In progress.

## Awards

GSBS/MDACC Student Travel Award
- Joint Meetings of 2025 Taipei International Statistical Symposium and the 13th ICSA International Conference, Taipei, Taiwan Oct 2025

GSBS Quantitative Sciences Program Travel Award
- 2025 Taipei International Statistical Symposium, Taipei, Taiwan Oct 2025
- American Society of Human Genetics Annual Meeting, Denver, CO Oct 2024

Grants for Young Scholars Attending International Academic Conferences Abroad, Academia Sinica Aug 2018

Poster Competition Award, Institute of Statistical Science, Academia Sinica Feb 2018

Academic Excellence Award (Ranking: 2/41), Department of Statistics, National Chengchi University, Spring 2015

## Presentations

SHARC: improved LoF SNV and Indel severity prediction and cross-type recalibration of coding variant annotation scores for clinical classification
- American Society of Human Genetics Annual Meeting, Montreal, Canada (Poster) Oct 2026

VACANT: A Tool for Annotation-Stratified Rare Variant Association Analysis and Effect Size Estimation
- International Genetic Epidemiology Society Annual Meeting, Estérel, Québec, Canada (Platform Presentation) Oct 2026

VACANT: An Adaptive Framework for Rare-Variant Association Testing that Leverages Continuous Functional Annotations

- Joint Meetings of 2025 Taipei International Statistical Symposium and the 13th ICSA International Conference, Taipei, Taiwan (Poster) Dec 2025

Annotation-based dynamic variant sets boost univariate and multivariate rare-variant association discovery in biobank cohorts

- American Society of Human Genetics Annual Meeting, Boston, MA (Poster) Oct 2025

VACANT-M: An Annotation-Enhanced Variant-Set Association Test Accounting for Genetic Relatedness and Population Structure

- International Genetic Epidemiology Society Annual Meeting, Cologne, Germany (Poster) Sep 2025

VACANT: Variant Annotation Clustering AssociatioN Test

- American Society of Human Genetics Annual Meeting, Denver, CO (Poster) Nov 2024

A meta-analysis approach to causal mediation modeling of semi-competing risks.

- The 28th South Taiwan Statistics Conference, National Chung Hsing University, Taichung, Taiwan, Jun 2019

Mediation Analysis with Causally Ordered Mediators Using Cox Proportional Hazards Model

- Royal Statistical Society 2018 International Conference, Cardiff City Hall, Wales, United Kingdom, Sep 2018
- The 27th South Taiwan Statistics Conference, National Cheng Kung University, Tainan, Taiwan, Jun 2018

MLB Database and Data Analysis

- Taiwan R User Group May 2018

## Teaching Experience

Teaching Assistant Sep 2015– Jun 2016
Calculus
Instructor: Yen-Lung Tsai, Ph.D,
Department of Mathematical Sciences, National Chengchi University

- Designed two-hour lectures and prepared exercises according to the syllabus every week.
- Helped students to finish GeoGebra assignments.

## Skills

Advanced Proficiency: R, Python, LaTeX
Statistical Genetics & Bioinformatics Tools: PLINK, REGENIE, SAIGE
Intermediate Proficiency: Matlab, SQL, Linux, C

## References

Chad D. Huff, Ph.D.
Professor
Department of Epidemiology
University of Texas MD Anderson Cancer Center, United States

Yen-Tsung Huang, MD, ScD
Research Fellow
Institute of Statistical Science
Academia Sinica, Taiwan

*[Updated on October 28, 2026]*